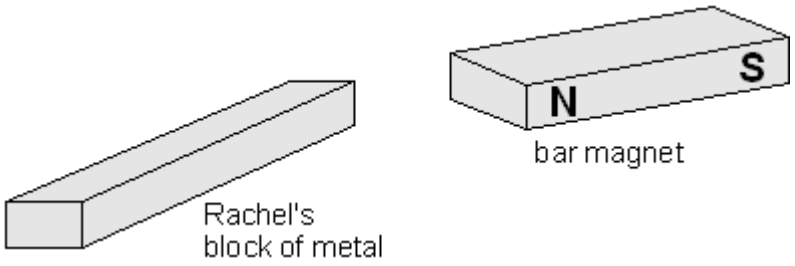
Q1. Rachel has a small block of metal. She wants to know if it is a magnet.



She holds the North pole of the bar magnet near one end of the block of metal. They attract. Then she holds the South pole of the bar magnet near the same end of the block of metal. Again they attract.

(a) Explain how this shows that Rachel's block of metal is **not** a permanent magnet.

.....

.....

1 mark

(b) What metal could Rachel's block be made from?

.....

1 mark

(c) Rachel puts the block of metal in a thin plastic bag.
What effect, if any, will the bar magnet now have on the block?

.....

.....

1 mark

(d) Rachel puts the bar magnet into a pile of coins. Some of the coins are attracted to the magnet but some are not.
Suggest a reason for this.

.....

.....

1 mark
Maximum 4 marks

Mark schemes

Q1.

(a) any **one** from

- a magnet would repel one end of the bar magnet
  'accept 'a magnet would be repelled by one end'
- it did not repel either end
- it attracted both poles
  accept 'a magnet would not attract both poles
  ***or** both ends'*

1 (L4)

(b) iron **or** steel **or** an alloy of iron
*accept 'nickel' **or** 'cobalt'*

1 (L4)

(c) any **one** from

- it would attract it
  accept 'attract'
- the same effect as before
  accept 'the bag would have no effect'
  ***or** 'the bag would make no difference'*
  ***or** 'no difference'*
  *do **not** accept 'no effect'*

1 (L4)

(d) any **one** from

- the coins which are attracted contain iron
  *accept 'some of the coins contain iron **or** steel'*
  ***or** 'some of the coins are made from a magnetic material'*
- some of the coins do not contain iron and some do
  accept 'some coins do not contain iron'
  *do **not** accept 'the coins are made from different metals'*

1 (L4)

[4]